Hints: Integrals – 2. Symmetry, the king's rule and periodicity

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The king's rule is the substitution $u = a + b - x$, which maps $[a, b]$ onto itself reversing orientation. For $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx$, call it I, apply the rule and add the result to I: the integrands add up to 1. For $\int_0^\pi x f(\sin x)\,dx$ use $u = \pi - x$ and $\sin(\pi - x) = \sin x$, then add.

Hint 2. $\int_0^\pi \frac{x \sin x}{1+\cos^2 x}dx = \frac{\pi}{2}\int_0^\pi \frac{\sin x}{1+\cos^2 x}dx = \frac{\pi}{2}\left[-\arctan\cos x\right]_0^\pi = \frac{\pi^2}{4}$. In the second integral, $\arctan x + \operatorname{arccot} x = \frac{\pi}{2}$ for $x > 0$, so it equals $\frac{\pi}{2}\ln\sqrt{3} = \frac{\pi \ln 3}{4}$.

Problem 2

Hint 1. All four are integrals over a full period (or a half period) of a function that is odd with respect to some point. Look for a substitution $x \mapsto c - x$ or $x \mapsto x + T$ that maps the interval onto itself (up to periodicity) and changes the sign of the integrand, or of a part of it.

Hint 2.

(a) $\arccos y = \frac{\pi}{2} - \arcsin y$, and $\arcsin(\sin x)$ is odd and 2π-periodic, so its integral over a period is 0; the answer is π^2. (b) $x \mapsto \pi - x$ (with periodicity) sends $\cos x \mapsto -\cos x$, and tan is odd: 0. (c) $x \mapsto 2\pi - x$ sends $\sin x - x \mapsto -(\sin x - x) - 2\pi$, and sin is odd and 2π-periodic: 0. (d) $x \mapsto \pi - x$ sends $x + \cos x \mapsto \pi - (x + \cos x)$ and $\cos(\pi - y) = -\cos y$: 0.

Problem 3

Hint 1. Split the interval into periods or into unit intervals on which the floor/ceiling functions are constant. (a) $|\sin 2018x|$ has period $\frac{\pi}{2018}$ and integral $\frac{2}{2018}$ over one period. (b) $x - \lfloor x \rfloor$ has period 1.

Hint 2.

(c) On $(n, n+1)$, $\lfloor x \rfloor \lceil x \rceil = n(n+1)$ and $\int_n^{n+1} x^2 dx = n^2 + n + \frac{1}{3}$, so each unit interval contributes $\frac{1}{3}$. (d) $\lfloor \log_2 x \rfloor = 5$ on $[50, 64)$ and 6 on $[64, 100]$.

Hint 3. Answers: (a) $2018^2 \cdot \frac{2}{2018} = 4036$; (b) $\frac{256}{3}$; (c) $\frac{2022}{3} = 674$; (d) $14 \cdot 5 + 36 \cdot 6 = 286$.

Problem 4

Hint 1. Over an interval that is a whole number of periods, $\int \sin mx \cos nx = 0$ always, and $\int \cos mx \cos nx = \int \sin mx \sin nx = 0$ for $m \neq n$ (positive integers). Expand the first product into four terms and keep only the one with equal frequencies.

Hint 2. First: only $\cos^2 3x$ survives, and $\int_{-2\pi}^{2\pi} \cos^2 3x\,dx = 2\pi$. Second and third: the integrands are odd and the interval is symmetric, so both are 0. Criterion: the integral vanishes if the integrand is odd about the centre of the interval, or if it is a product of trigonometric functions with different integer frequencies (or a $\sin \cdot \cos$ pair) over a full period.

Problem 5

Hint 1. Under $x \mapsto 1/x$: $dx \mapsto -dx/x^2$ and $\frac{dx}{1+x^2}$ is invariant, while $\ln x \mapsto -\ln x$. So $\int_0^\infty \frac{\ln x}{1+x^2}dx$ equals its own negative: 0. For the second identity, applying $x \mapsto 1/x$ to $\frac{1}{(1+x^2)(1+x^a)}$ gives $\frac{x^a}{(1+x^2)(1+x^a)}$; add the two forms to get $\int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$.

Hint 2. So the third integral is $\frac{\pi}{4}$ (take $a = \pi$). For $\int_0^\infty \frac{\ln x}{(1+x^2)^2}dx$ the inversion alone gives no information; substitute $x = \tan\theta$ to get $\int_0^{\pi/2} \cos^2\theta \ln\tan\theta \, d\theta$.

Hint 3. Write $\cos^2\theta = \frac{1+\cos 2\theta}{2}$; the $\frac{1}{2}$ part gives 0 (it is the first integral again), and $\frac{1}{2}\int_0^{\pi/2} \cos 2\theta \ln\tan\theta \, d\theta$ integrates by parts with $v = \frac{\sin 2\theta}{2}$, using $(\ln\tan\theta)' = \frac{1}{\sin\theta\cos\theta}$. The answer is $-\frac{\pi}{4}$.